What is cybersecurity? Explain the CIA triad with one real-world example for each part.

Cybersecurity is the practice of protecting computers, networks, servers, applications, digital devices, and sensitive information from unauthorized access, cyberattacks, damage, theft, and disruption.

It involves technologies, processes, policies, and security practices designed to protect digital systems and maintain the confidentiality, integrity, and availability of information.

Example:
A bank uses firewalls, encryption, multifactor authentication (MFA), and intrusion detection systems to protect customers' accounts and financial transactions.

CIA Triad

The **CIA Triad** is a fundamental cybersecurity model consisting of three security principles:

The three pillars of information security are **Confidentiality**, **Integrity** and **Availability**.

A. Confidentiality

Confidentiality ensures that sensitive information is accessible only to authorized users.

- Prevents unauthorized disclosure of information.
- Uses encryption, passwords, access controls, and MFA.

Real-world example:
In online banking, only the account holder and authorized bank employees can access confidential account details.

B. Integrity

Integrity ensures that data remains accurate, complete, and protected against unauthorized modification.

- Prevents unauthorized changes to information.
- Uses hashing, digital signatures, access controls, and audit logs.

Real-world example:

When a customer transfers ₹5,000, the banking system must ensure that the transaction amount is not maliciously changed to ₹50,000.

C. Availability

Availability ensures that systems, networks, and information remain accessible to authorized users whenever required.

- Protects systems against downtime and service disruptions.
- Uses backups, redundancy, failover systems, and DDoS protection.

Real-world example:

An e-commerce website uses load balancing and backup servers to remain accessible during high traffic.

Principle	Main purpose	Common security measure
Confidentiality	Prevent data disclosure	Encryption
Integrity	Prevent unauthorized modification	Hashing and digital signatures
Availability	Ensure reliable access	Backups and redundancy